



AIDS:UG2(PC)

Duration: 90 Mins (11:00AM-1:00PM)

Date: 20-02-2026

Max. Marks: 30

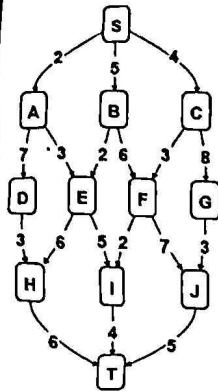
Instructions:

Roll No: _____

1. This is a **closed-book, closed-notes** examination.
2. Please **write/draw legibly**. If we cannot understand what you have written, we cannot grade it.
3. All answers must be written **in pen**. Answers written in **pencil will not be evaluated**.
4. **Mobile phones and laptops are strictly not allowed** inside the examination hall.
5. You may use a **scientific calculator**, if required.
6. Clearly mention the **question number** for each answer. Answers without proper question numbers may not be evaluated.

1	a. What do you mean by Rationality and Who is an Rational Agent in AI	[1 Mark]
	b. During the class, you had designed an agent, specify what are its task environment	[2 Mark]
	c. Using an example give differences between stochastic and deterministic type of environment	[2 Marks]
	d. Describe an Utility-based agent	[3 Marks]
2	a. Mention the steps involved in simple hill climbing algorithm and its advantages and limitations	[4 Marks]
3	Apply Genetic Algorithm to maximize: $f(x)=x(1-x)$ Subject to: $0 \leq x \leq 1$ Using 5-bit binary encoding and Population size = 8, with Maximum generations = 10. Given crossover Probability as 0.6 and mutation probability as 0.06. Perform one complete generation and determine the optimal value of x .	[8 Mark]

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Node	$h(n)$
S	10
A	9
B	6
C	7
D	6
E	5
F	4
G	3
H	6
I	2
J	4
T	0

[10 Marks]

Consider the above weighted graph, numbers on edges represent the cost between the two nodes and the table aside represents the heuristic cost to reach the goal node. Start node: S and Goal node: T . Find a path from S to T .

1. Perform BFS and show expansion order.
2. Perform DFS and show path obtained.
3. Apply A* Search
 - a. Show $g(n)$, $h(n)$, $f(n)$
 - b. Maintain stack and fringe
4. Compare:
 - a. Path found by BFS, DFS and A*
5. Comment on heuristic effectiveness.